

5.1. Human Robot Interaction Challenge

Description

The robot is working as a receptionist at a party. The robot has to take two new guests to the living room, offer a free place to sit and introduce them to each other. The second guest brings a bag for the robot to carry.

Main goal: The robot welcomes and assists two newcomers at a party and offers them a seat, while maintaining appropriate gaze direction during conversation. The second guest brings a bag to be placed somewhere in the house.

Additional goals:

1. Detect doorbell sound as a signal that a guest has arrived.
2. Open the entrance door for each arriving guest.
3. Visually describe the first guest to the second guest.
4. Understand each guest's name and favorite drink without asking any confirmation or correction questions (non-essential questions).

Focus

System Integration, Human-Robot Interaction, Person Detection, Person Recognition

Setup

- **Location:**

- The test takes place in the living room.
- The robot starts inside the *Arena* at a predefined location near where guests are sitting.
- **Entrance:** The entrance door is open by default. The team leader can request it to be closed to allow the robot to earn additional points by opening it for the guests.

- **People:**

- **Arriving Guests:** Both guests have a name and a favorite drink. An arriving guest will wait by the door, and the robot has to move to meet guests by the door. A guest will not move inside the house until the robot has initiated conversation. It is not allowed to instruct guests to move in front of the robot to recognize the guest and start the conversation. The robot is the one that must approach the guest. Guests have to be guided to the living room where the robot will offer a free place to sit and introduce the guests to each other. Each guest will arrive separately.
- **Passive Guests:** A few individuals, including possibly spectators, can be anywhere in the arena — sitting or standing in relevant places, e.g., on the couch, near the seats, or in the area with the drinks.

Procedure

Both guests arrive separately. A doorbell will ring to signal the guest has arrived. The doorbell will only ring if the robot is waiting for a new guest at the starting position. The robot either

opens the door for the guest or meets them at the door. It greets the guest and asks for their name and favorite drink. The robot escorts the guests to the living room and offers a free seat. Once both guests are seated, the robot introduces them to each other. To introduce the two guests, the robot must (for both guests) look at a guest and state the other arriving guest name and favorite drink. The second guest will bring a bag for the host. The robot should ask for the bag to be handed over and move to the host. The host is already waiting and instructs the robot to follow him/her. The host will move to a random place in the house and inform the robot it can drop the bag there.

- **Doorbell:** A doorbell sound will be played when a guest is by the door ready to be met. The doorbell sound will not be provided beforehand and may be any traditional doorbell sound. As an alternative, if no doorbell sound is available, the guest can knock on the door. Waiting for the doorbell is optional, robots can bypass it and directly meet the guest at the door. In order for the doorbell sound to be considered detected, the robot must be waiting at the starting position and react instantly to the sound.
- **Greeting guests:** The names of the guests can be any popular english names. The favorite drinks of the guests can be any english named popular drink (it may not be in the list of objects provided). The guest paces slightly to the left and right during the conversation with the robot. Other people might appear in the background.
- **Looking at person:** The robot must look at the person it is talking to. Guests will move slightly to confirm the robot dynamically follows them with its gaze. Points for gaze tracking are awarded only if the robot continuously tracks the moving person.
- **Looking at direction of navigation:** The robot must look in the direction it is navigating. Persistently looking in an unrelated direction or at unrelated people during navigation results in a score deduction.
- **Seating People:** The robot must clearly indicate or point to a location or seat for each guest.
- **Switching Places:** After being seated, guests may switch seats.
- **Introductions:** When introducing guests, the robot must clearly identify the person being introduced and state their name and favorite drink. The robot must be looking at a guest and state the other arriving guest name and favorite drink. The robot must do this for both guests. The Robot is allowed to break eye contact if it is natural to do so, e.g. looking at the guest while stating their name or indicating a free seat. Introducing two people means to introduce them to each other.
- **Non-Essential Questions:** If the robot does not ask any non-essential questions to help understand, confirm, or correct information, bonus points are awarded. Questions such as "Did you say your name is James?" or "Is your favorite drink milk?" are considered non-essential questions. However, asking a guest to repeat an answer because the robot could not hear is not considered a non-essential question.
- **Bag handover:** The robot must grab the bag from the second guest hand. If this is not possible, the robot can instruct the guest to place the bag somewhere in its structure
- **Follow the host:** The host will be sitting in the living room. The robot must inform the host that it has a bag and ask the host for guidance. When requested by the robot, the host will stand in front of the robot. When the robot is ready to follow, it should inform the host. The host then walks naturally toward the designated bag drop location. Once at the correct location, the host informs the robot that it should place the bag on the floor.

- **Not acknowledging people:** If the robot fails to attend to a person attempting to interact with it, e.g., by repeatedly looking away, not orienting towards the speaker, or focusing on unrelated objects during interaction.

Additional rules and remarks

1. **Misunderstanding:** Not understanding the guests and asking them again is fine. Continuing with a wrong name or favorite drink causes a score reduction.
2. **Partial Scoring:** The main task allows partial scoring per guest.
3. **Additional Seating:** Referees may add chairs to the living room area.
4. **Human Assistance:** Score reduction applies per guest as follows:
 - **Custom Operator:** Since the main focus of the test is HRI, no custom operator can be chosen.
 - **Alternative HRI:** Using an alternative HRI to understand a guest causes a score reduction per information (name or favorite drink).
 - **Recognizing People:** If the robot has to ask for help to identify people, the score is reduced.
 - **Follow the host:** Penalties apply if the robot loses the operator and needs help recovering. The following forms of assistance are penalized:
 - 4.1. Natural interaction (e.g., waving and calling)
 - 4.2. Unnatural interaction (e.g., raising both hands and jumping)
 - 4.3. Returning in front of the robot to resume following.
 - 4.4. Physical contact (e.g., pulling the robot's hand)
5. **Bag handover:** If the robot has to ask the guest for help during handover, the score is reduced. If the robot tells the guest to place the bag somewhere on its structure, no points are awarded. To score the maximum amount of points:
 - The robot must clearly signal its readiness to receive the item (e.g., by holding its manipulator still, using lights, audio cues, or verbal prompts).
 - The manipulator must be positioned at least within 10 cm of the operator's hand.
 - The operator can complete the handover; the robot should not reach, grasp, or initiate dangerous movement toward the human.
 - Explicit verbal guidance from the operator (e.g., *"Place the item in my gripper, I'll close it in 3...2...1"*) will result in penalties.
 - The robot may, however, issue safety cues such as: *"I detect the object in my hand. I will close my gripper in 3 seconds, please be careful."*

Instructions

To Referee

The referees need to:

- Assign name and favorite drink to two volunteers.
- Arrange and rearrange people in the living room.
- Open the entrance door upon request from the robot.

To OC

During setup day:

- Announce the bag that will be used for handover.

At least two hours before test:

- Announce the starting position.
- Recruit four volunteers: two as arriving guests and two as passive guests.

Score sheet

The maximum time for this test is **7:00** minutes.

Action	Score
<i>Greeting</i>	
Detect the doorbell sound as a signal that a guest has arrived	2×30
Open the entrance door for a guest	2×200
Look at the person talking, when receiving a guest	2×50
Offer a free seat to the new guest	2×100
Look in the direction of navigation or at the navigation goal	2×15
Tell a visual attribute of the first guest to the second guest	
Correct attribute	$4 \times +20$
Incorrect attribute	4×-20
Not asking non-essential questions to confirm or correct information	4×15
<i>Introduction</i>	
Say name and favorite drink of each guest (during introduction)	4×30
While introducing guests, look to the correct guest while talking about the other guest	2×50
<i>Guiding</i>	
Receive the bag from the guest via handover	50
Requesting handover assistance from the guest	-25
Bag placed on the robot structure instead of being handed over	-50
Following the host to the bag drop area	200
Drop the bag while following the host	-50
Rediscovering the operator by natural interaction	-50
Asking the operator to wait	-50
Guiding the robot with physical contact (take by the hand)	-150
Drop the bag in the correct area	50
<i>Penalties</i>	
Wrong guest information was memorized	4×-40
Alternative HRI	6×-20
Not acknowledging people	2×-200
<i>Special Penalties & Bonuses</i>	
Not attending (see sec. 3.8.1)	-500
Outstanding performance (see sec. 3.8.3)	145
Total Score (excluding special penalties & standard bonuses)	1450